Omada Port Profiles

Trunk_all N/U - 99 T - 10,20,30,40

Trunk_AP N/U - 99 T - 10,30,40

Trunk_Proxmox - N/U - 99 T - 10,20

MGMT_Access - N/U - 10 T - N/A

SRV_Access - N/U - 20 T - N/A

Trusted_Access - N/U - 30 T - N/A

IOT/Guest_Access - N/U - 40 T - N/A

Reservations
All start with 192.168
And gateways for each vlan are VLAN.1

Vlan 10

Router 10.1
Controller 10.2
Main SW 10.3
SFP+ Agg SW 10.4
Access SW 10.5
AP1 10.6
AP2 10.7

Vlan 20

Proxmox 20.10
Pihole1 20.11
Pihole2 20.12
Gameserver 20.50
NAS 20.20
Frigate 20.30
Home Assistent 20.40

VLAN 30

AdminPC 30.50
AdminPhone 30.51
AdminLaptop 30.52

VLAN 40

Laser 40.2
Inkjet 40.3
3DPrinter 40.4

VLAN 99 (Throwaway Native) - Used for trunk ports to prevent VLA hopping.

VLAN 10 - Management/Network - 192.168.10.0/24
Includes - Router, Switches, Controller, WAN

VLAN 20 - Servers/Resources - 192.168.20.0/24
Includes - TrueNAS Server, Proxmox Server, Proxmox VM's/Containers(Also PIhole 1 and 2 and game server).

VLAN 30 - DHCP Clients/Trusted Users - 192.168.30.0/24
Includes - Main WiFi, Main Ethernet, DHCP Clients, Personal Devi (Must allow management/Network and Servers/Resources from personal PC).

VLAN 40 - IOT Devices/Guests - 192.168.40.0/24
Includes - IOT Devices, Cameras, Guest Wifi, Printers.

The VLAN Interfaces in OPNsense are the VLAN ID Dot 1 (Examp 192.168.3.1.)

All VLAN's have DHCP ranges of 101-254. Reservations will be us for static devices/mappings.

All devices except network gear and servers should use the 2 piho servers.

Network Gear should use Unbound directly within their ip ranges gateway. EX.(192.168.30.1) Not the management router of 192.168.10.1 unless its on vlan 10.

Ports with only a port number are disabled, will be configured later when needed.

Ports without anything plugged in are disabled by default unless otherwise needed.

AN

ices

ble -

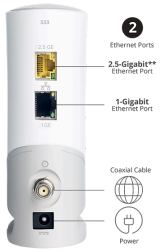
sed

ble



Proxmox Server
192.168.20.10/24

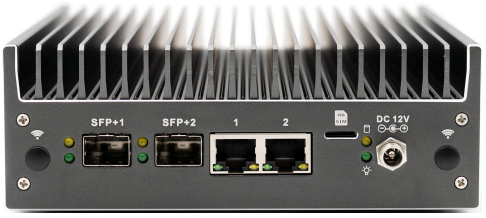
Pi-hole 1 - 192.168.20.11/24
Pi-hole 2 - 192.168.20.12/24
Game Server 1 - 192.168.20.50
Frigate - 192.168.20.30
Home Assistant - 192.168.20.40
(Unbound running on the router is used for upstream).



Internet/Wan
PUBLIC IP
xxx.xxx.xxx.xxx/xx



TrueNAS Server
192.168.20.20/24



Opnsense Router
192.168.10.1/24
Running DHCP Server and Local Unbound DNS.
(All other devices and DHCP settings should use PIhole IP).



Fallback Admin PC
192.168.99.2
(Connected directly to router with gatway of .1
(Should have access to all VLAN'S)



Omada Controller
192.168.10.2/24



Omada Main Switch
192.168.10.3/24



Omada SFP+ Agg Switch
192.168.10.4





Omada Access Switch
192.168.10.5



Omada AP 1
192.168.10.6/24

SSID - Main - VLAN 30
SSID - IoT - VLAN 40

Omada AP 2
192.168.10.7/24

SSID - Main - VLAN 30
SSID - Iot - VLAN 40



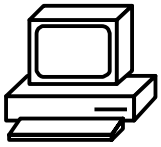
Personal Admin PC
192.168.30.50
(Should have access to all VLAN'S, N/U - 30)



Personal Admin Phone
192.168.30.51
(Should have access to all VLAN'S)
(Connected to Main WIFI)



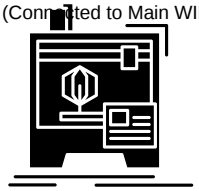
Personal Admin Laptop
192.168.30.52
(Should have access to all VLAN'S, N/U - 30)



Client PC
DHCP Assigned - VLAN 30



Laser Printer
192.168.40.2
(Connected to IOT WIFI)

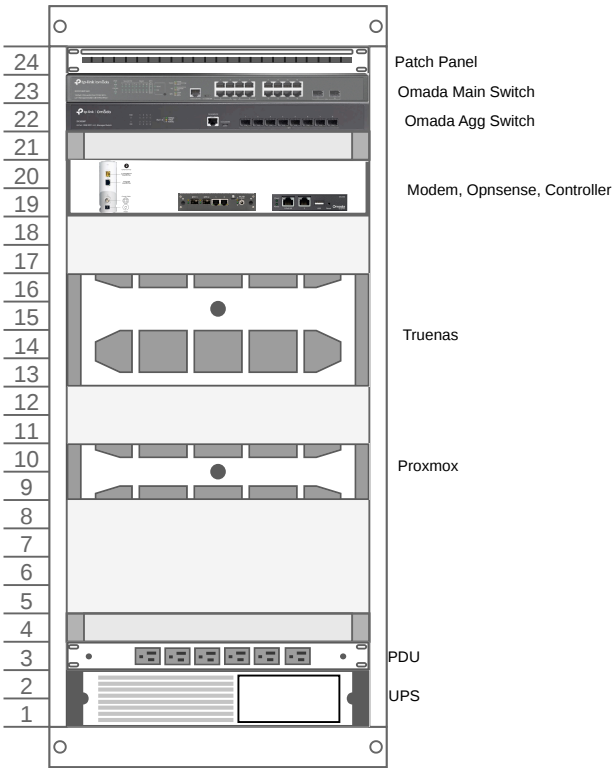


3d Printer
192.168.40.4
(Connected to IOT WIFI)



Inkjet Printer
192.168.40.3
(Connected to IOT Wifi)

3DPrinter 40.4
Doorcam 40.5
Garagecam 40.6
Backcam 40.7





AP1



AP2



Access Switch

Devic

Main

Main

SFP-

Acces

Contr

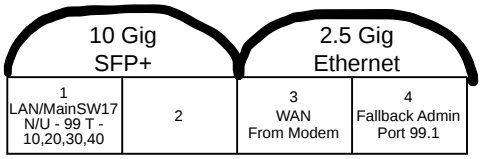
AP1

AP2

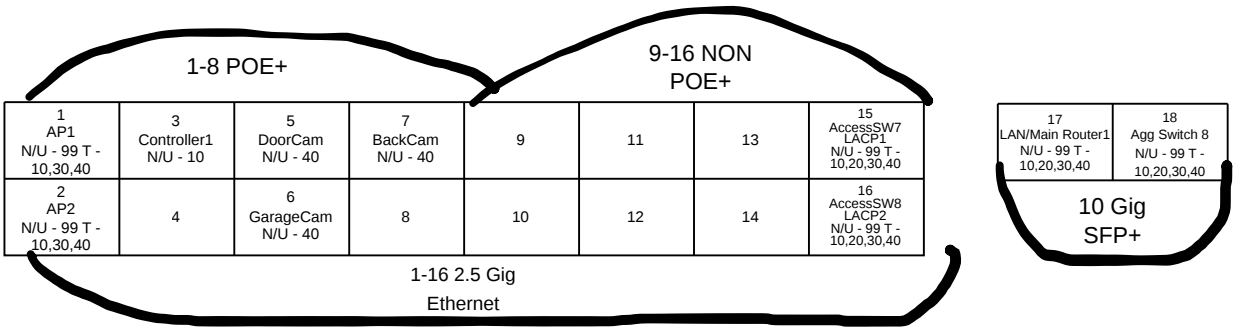
Proxm

ce Port Mapping/VLAN Tagging

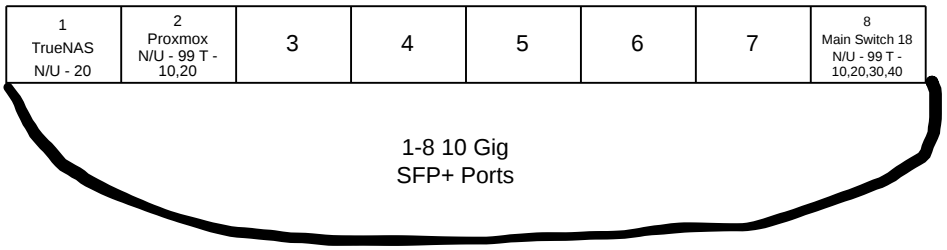
Router



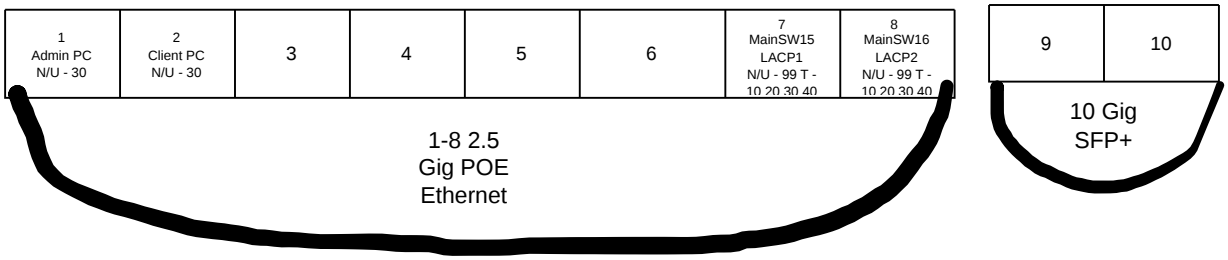
Switch



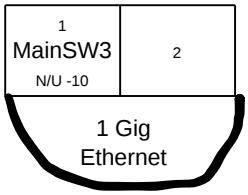
+ Agg Switch

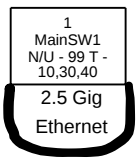


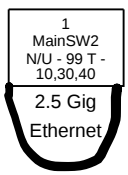
ss Switch



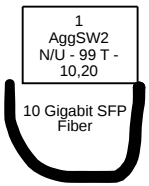
roller

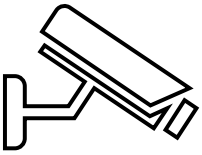




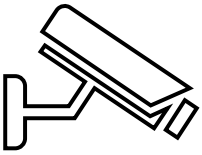


mox Server

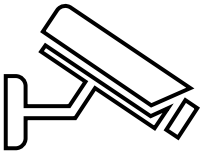




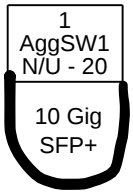
DoorCam
192.168.40.5
(POE To Main SW)



GarageCam
192.168.40.6
(POE To Main SW)



BackCAM
192.168.40.7
(POE To Main SW)


1
AggSW1
N/U - 20
10 Gig
SFP+